

J/ψ radiative corrections — status report

16–17 September 2026

1 The final formula is missing from the NLO paper

The NLO paper gives δ_{self} (8), δ_{vert} (12), δ_{soft} (13), δ_{hard} only as an unevaluated integral (14), and the exact Bardin–Shumeiko result (28). **There is no compact closed form for the total ultrarelativistic correction** — the quantity an experiment actually applies. The right one is Eq. (24)/(26) of Ehlotzky & Mitter with the missing term restored:

$$\delta(m_V, \omega_{cut}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_\ell} \right) \left(\ln r + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r) + \underbrace{\frac{r^2}{16} - \frac{3}{8}r}_{\text{missing from [10]}} \right]$$

$$r = \frac{2\omega_{cut}}{m_V}, \quad R(r) = \left(r - \frac{r^2}{4} - \frac{3}{4} \right) \left[\ln \left(\frac{m_V}{m_\ell} \sqrt{1-r} \right) - \frac{1}{2} \right], \quad L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt$$

For muons add the electron-loop vacuum polarization = δ_{self} of Eq. (8) with m_e . Equivalently the missing piece is $+(\alpha/\pi)(3r/2 - r^2/4)$ added to δ ; it vanishes as $r \rightarrow 0$ and reaches $5\alpha/4\pi = 0.0029$ at $r = 1$.

Status of this formula. It is *not* exact in the lepton mass, and not the $m_\ell = 0$ limit either: the mass is kept in the logarithms, while terms of relative order $(m_\ell/E)^2$, $E = m_V/2$, are dropped. Ehlotzky & Mitter say so above their Eq. (22) (“neglecting terms of order $(m/E)^2$ ”). Numerically $(m_e/E)^2 = 1.1 \times 10^{-7}$ and $(m_\mu/E)^2 = 4.7 \times 10^{-3}$, so at $r_{max} = 1$ it is off by 1×10^{-7} for electrons and -7×10^{-5} for muons, against the exact Eq. (28).

Where it should go: Section IV

The formula belongs in Section IV of the NLO draft, the ultrarelativistic section, immediately after their Eq. (14) — the closed form *is* the evaluation of that integral combined with Eqs. (8), (12) and (13). Three arguments, all from their own text:

- Section IV opens by promising “compact expressions for all contributions to decay width: both for virtual and real bremsstrahlung ones.” Compact expressions follow for every individual contribution — but not for the sum, which is the only physical combination.
- Their Eq. (16), $\delta_{FSIC} = 3\alpha/4\pi$, becomes an immediate corollary: it is the $r_{max} \rightarrow 1$ limit of the closed form, less δ_{self} . At present it arrives as a separate assertion.
- Their Figure 4 plots exactly this quantity for both channels. The reader is shown the curve but never given the function that draws it.

Section IV also states that the URA is “applicable for $\ell = e$ ” and that the mass-dependent treatment is “crucial for ...the muon channel”. That matches the accuracy table above. So the electron

expression should be presented as the working formula of Section IV, and the muon expression alongside it with the $(m_\mu/E)^2$ error flagged and Eq. (28) named as the fallback. Dropping the muon formula in unqualified would contradict a sentence they have already written two pages earlier.

This suggestion, with the formulas written out for both channels, is Section 5 of `NLO_comments.pdf`.

2 The missing term is Ehlotzky's error, not yours

Your Eqs. (14), (15), (19), (20) are *faithful transcriptions* of Ehlotzky & Mitter (E&M) Eqs. (24) and (26):

$$\begin{array}{cccc} \text{VPK (11)} \leftrightarrow \text{E\&M (22)} & \text{VPK (13)} \leftrightarrow \text{E\&M (23)} & \text{VPK (14)} \leftrightarrow \text{E\&M (24)} & \text{VPK (18)} \leftrightarrow \text{E\&M (25)} \\ & & \text{VPK (19)} \leftrightarrow \text{E\&M (26)} & \end{array}$$

E&M Eq. (23), the differential spectrum, is correct. The term was lost when that spectrum was integrated analytically in 1968. Their Eq. (26) and their Table I inherit it. Recomputing their Table I from their own Eq. (26) reproduces **every printed digit**, electrons and muons:

r_{max}	E&M δ	Eq. (26) as published	corrected
0.10	-0.087	-0.087	-0.087
0.25	-0.034	-0.034	-0.033
0.50	-0.002	-0.002	+0.000
1.00	+0.021	+0.021	+0.024

The error has stood for 57 years. Your paper is the first to correct it.

3 Table I of the NLO paper: $m_V = 1.5$ GeV is correct

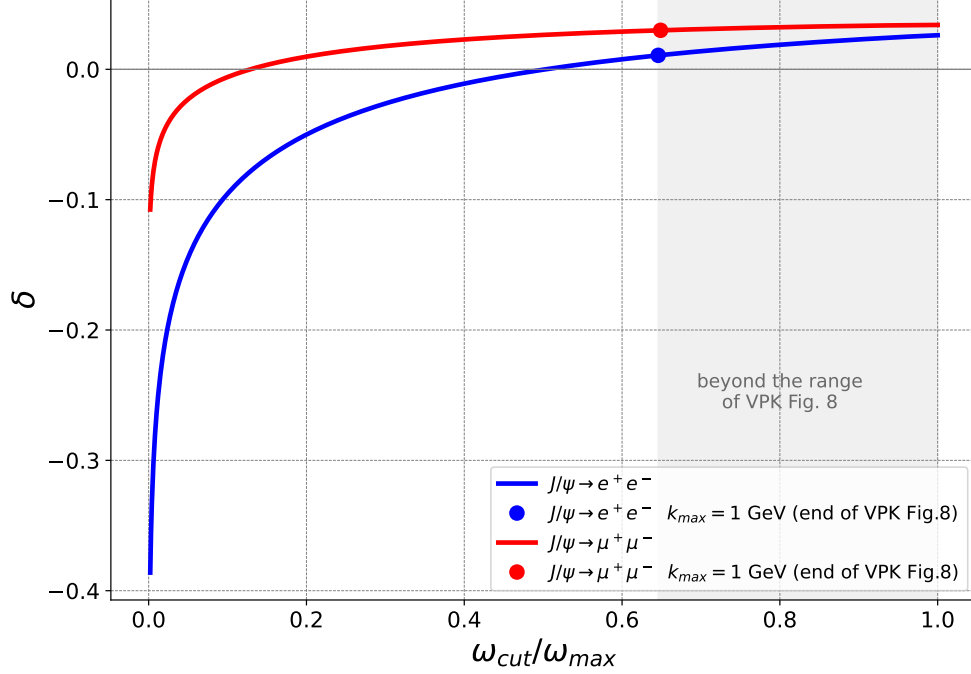
You suspected it was wrong; it is not. E&M's table column is $2E$ (MeV), and $2E = m_V$ because $E = m_V/2$ at resonance (stated below their Eq. 22). The rows are marked ρ at 765.0, ω at 782.0, ϕ at 1019.5 — meson masses. So the last row, 1500.0, is $m_V = 1.5$ GeV, exactly what the NLO paper used. δ_R settles it, depending on the mass only through $\ln(2E/m)$:

reading	δ_R		
$m_V = 1.5$ GeV (NLO's choice)	-0.746525	$\leftarrow$ E&M print	-0.747
$m_V = 3.0$ GeV (if 1500 were E)			-0.868832
$m_V = M(J/\psi)$			-0.874655

What *is* wrong is their claim of agreement with that table: at $r_{max} = 1$ they have 0.023891 against E&M's 0.021. They agree only in the small- r_{max} rows.

4 NLO Fig. 4 vs VPK Fig. 8

Same quantity, different axes.



(a) **The endpoints differ.** NLO uses $\omega_{max} = m_V/2 - 2m_\ell^2/m_V$ (their Eq. 15); you use $k_{max} = M_V/2$. Identical for electrons, but for muons 1.54125 vs 1.54846 GeV, i.e. $r_{max}^{max} = 0.99535$, not 1 — a 0.47% shift.

(b) **Your Fig. 8 covers only the first two thirds**, stopping at $k_{max} = 1.0$ GeV = $0.646\omega_{max}$. Their Fig. 4 runs to 1.

$\omega_{cut}/\omega_{max}$	$k_{max}(e)$	$\delta(e)$	$k_{max}(\mu)$	$\delta(\mu)$
0.10	0.1548	-0.095444	0.1541	-0.005934
0.25	0.3871	-0.036636	0.3853	+0.014282
0.50	0.7742	-0.000275	0.7706	+0.026393
1.00	1.5485	+0.026135	1.5412	+0.034023

At the endpoint against their exact Bardin–Shumeiko numbers: e^+e^- yours 0.026135 vs exact 0.026135 (diff 10^{-7}); $\mu^+\mu^-$ yours 0.034017 vs exact 0.034097 (diff -8×10^{-5} , the expected URA limit).

5 New finding: your $k_{max} = M_V/2$ contradicts your own Eq. (22)

The text above your Eq. (16) says the maximum photon energy is $k_{max} = M_V/2$. That is the massless limit. The exact endpoint is

$$\omega_{max} = \frac{M_V^2 - 4m_\ell^2}{2M_V} = \frac{M_V}{2} - \frac{2m_\ell^2}{M_V},$$

which is NLO Eq. (15) — **and it is already in your paper**, as Eq. (22), $E_\gamma = (M_{J/\psi}^2 - M_{ee}^2)/(2M_{J/\psi})$ evaluated at $M_{ee} \rightarrow 2m_\ell$. So Eqs. (16) and (22) of your own paper disagree. No consequence for electrons. For muons r_{max} can only reach 0.99535, so the $r_{max} = 1$ row of a muon table lies outside phase space. **Not fixed — your call.**

6 Formula-by-formula status

VPK	NLO	verdict
(1)–(5)	(3)–(5)	agree; conventions differ (fixed λ vs polarization-averaged)
(6)	(1)+(2)	agree exactly. Fixed today: $4m_\ell^2/m_V \rightarrow 4m_\ell^2/m_V^2$
(7)	(6)	identical (sympy: difference 0)
(8)	—	ultrarelativistic limit; NLO has none
(11),(12)	(8)+(12)+(13)	identical to machine precision
(13)	(14) integrand	identical
(14),(15)	—	were missing the term. Fixed today
(16)	(15)	disagree — see Section 5
(19),(20)	—	were missing the term. Fixed today; Eq. (20) $m_e \rightarrow m_\mu$ also fixed
(21),(22)	—	no counterpart; (22) is equivalent to NLO (15)
—	(16)	FSIC $3\alpha/4\pi$; your corrected Eq. (19) reproduces it
—	(28)	exact Bardin–Shumeiko; no counterpart in your paper

7 Changed today

Paper (`Jpsi_RC.tex`, compiled and published): Eqs. (14),(15),(19),(20) missing term added; Eq. (20) leading $\log m_e \rightarrow m_\mu$; Eq. (6) $4m_\ell^2/m_V \rightarrow 4m_\ell^2/m_V^2$; Fig. 7 recomputed at $\varepsilon = 0.001$ (was 0.01, caption said 0.001); Figs. 3–8 in-plot titles removed, information moved to captions; Figs. 3,4,6 captions now name the two decay modes.

Code: three duplicate copies collapsed into one library `rc_vector_mesons.py`; `validate_rc.py` checks it against an independent quadrature reference and two analytic anchors — all pass. Repo `~/jpsi-rc-2026`, <https://github.com/vkubarovsky/jpsi-rc-2026>, mirrored to OneDrive Work/2026_Jpsi_rad

8 Waiting on you

- `notes/NLO_comments.pdf` — the note for your co-authors: all six of your items plus the Eq. (4) polarization-sum point. *Not sent anywhere.*
- Section 5: Eq. (16) vs Eq. (22) inconsistency — not fixed.
- “Formala” $\times 2$ in the Fig. 9 legend — you said leave Fig. 9 alone.
- `fig_igor` defined twice in `Jpsi_rad_corr.py`.
- Never verified: your Eqs. (2)–(5) in detail, (17), (18); and the Bethe–Heitler formulas behind the -20% claim in Section VII.